

Warm-up: Representing Real-World Data

As of 2020, the concentration of carbon dioxide in Earth's atmosphere is at nearly 412 parts per million (ppm) and rising. This represents a 30 percent increase since 1960, when the concentration was near 315 ppm, and an 11 percent increase since 2000, when it was near 370 ppm.^a

^a<https://climate.nasa.gov/news/2915/the-atmosphere-getting-a-handle-on-carbon-dioxide/>

Discussion Question

- What might be some other ways to express the data given?
Come up with at least two.
- What questions might you want to investigate, based on this data?

Lecture 2 Real-World Functions

Big Questions

- What are some useful ways to represent real-world data?

Warm-up: Representing Real-World Data

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Discussion Question

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- What questions might you want to investigate, based on this data?

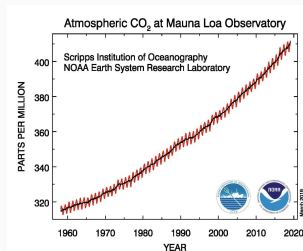
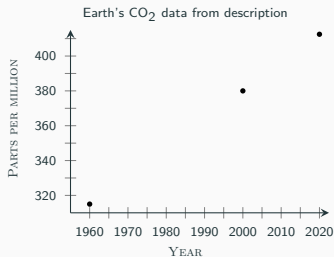
Sample Representations

There is no unique correct answer. You might have come up with others!

- Let $C = f(t)$, where t is the time in years since 1960 and C represents the concentration of carbon dioxide in Earth's atmosphere in parts per million (ppm) at time t . Then:

$$f(0) \approx 315, \quad f(40) \approx 370, \quad f(60) \approx 412.$$

- It is also possible to graph this data.



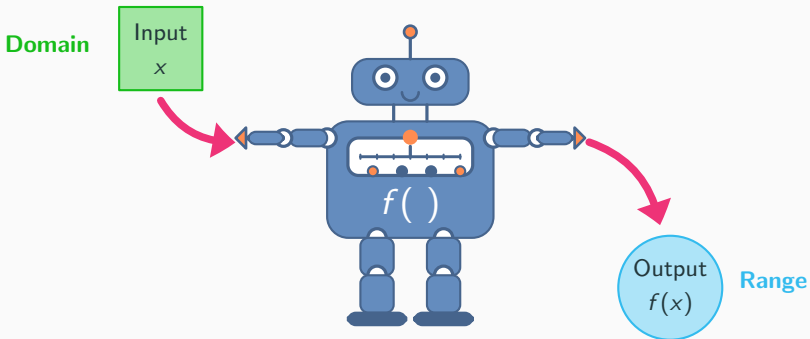
What is a function?

Definition

A **function** is a rule that takes certain numbers as inputs and assigns to each a definite output number.

The set of input numbers is called the of the function.

The resulting set of output numbers is called the of the function.



Think By Yourself

On your own, take one minute of **quiet time** to:

- Think of an example of a function.
- Identify your function's domain.
- Identify your function's range.

Examples of Functions

Discussion Question

As a group, share your examples, writing down at least four examples of functions. For each, what is the domain? What is the range?

Function	Domain	Range

More Examples of Functions

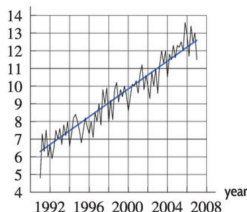
A bank account balance (\$) as a function of time t (in years):

$$f(t) = 500e^{0.03t}.$$

The mass of cat food (in grams) in Lillibet's bowl as a function of time t (in minutes since 8:00am).

US oil imports as a function of time.

US oil imports
(million barrels per day)



The percentage of people, P , below poverty level in the US t years since 2015:

t	0	1	2	3
P	13.5	12.9	12.3	11.7

Four Representations of Functions

We consider four main representations of functions:

1. **Graphs**
2. **Tables**
3. **Formulas**
4. **Verbal descriptions**

After Class

Update your own examples to include one function given in each representation!

Advantages and Disadvantages of Representations

Not all representations are equal!

- Each representation has different strengths and weaknesses.
- It is important to keep these in mind when communicating your ideas.

Function Representations: Tables of Data

Total revenue generated by arcades in the US (from FRED)

year	2000	2001	2002	2003	2004	2005	2006
billion USD	1.196	1.176	1.269	1.240	1.307	1.435	1.601

Discussion Question

What are the advantages of such a representation? disadvantages?

Function Representations: Formulas

The amount of a drug in the patient's blood stream is modelled by

$$C = 2te^{-\frac{t}{2}} \text{ ng/mL (nanograms per milliliter)}$$

where t is in hours (since taking the drug).

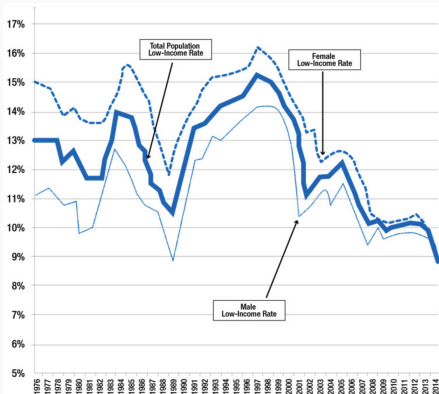
Discussion Question

What are the advantages of such a representation? disadvantages?

Function Representations: Graphs

Discussion Question

What are the advantages of such a representation? disadvantages?



From *A Backgrounder on Poverty in Canada*.

Function Representations: Words

“The composition of Earth’s atmosphere has most certainly been altered. Half of the increase in atmospheric carbon dioxide concentrations in the last 300 years has occurred since 1980, and one quarter of it since 2000.”

Discussion Question

What are the advantages of such a representation? disadvantages?

Value of a Car

The value of a car decreases as the car gets older. Let's consider the value of this car, V , in thousands of dollars, as a function of the age of a car, a , in years: $V = f(a)$. The car owner hasn't taken a calculus course before.

Poll Question

Which statement most accurately captures the plain english meaning of " $f(5) = 6$ "?

- A** The car is worth 5 thousand dollars less after 6 years.
- B** The car is worth 5 when it is 6 years old.
- C** The car is worth \$5,000 when it is 6 years old.
- D** The car is worth 6 thousand dollars when it is 5 years old.

Value of a Car

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Discussion Question

How would you explain the meaning of a **vertical intercept** of 15 to the owner? What about a **horizontal intercept** of 10?

After Class

Try drawing a possible graph of the function $V = f(a)$. Label each axis as V or a and to include units!

Be sure to:

- **Sign in to WileyPLUS** using your mail.utoronto.ca email address, if you haven't already done so.

You may need to wait 1-2 business days for an email from WileyPLUS if:

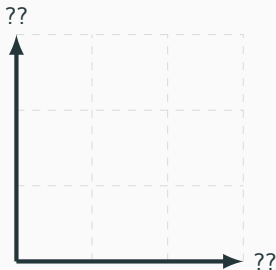
- You've recently enrolled in the course.
- You haven't activated your UTORid.
- **Read Section 1.1** and review the definition of a **linear function** in Section 1.2.
- Complete Problems 1-6 of **Homework 1**.
- Work on your Week 1 **Reflection!**

Have a great day!

For Extra Practice

"As I drove down the highway this morning, at first traffic was fast and uncongested, then it crept nearly bumper-to-bumper until we passed an accident, after which traffic flow went back to normal until I exited."

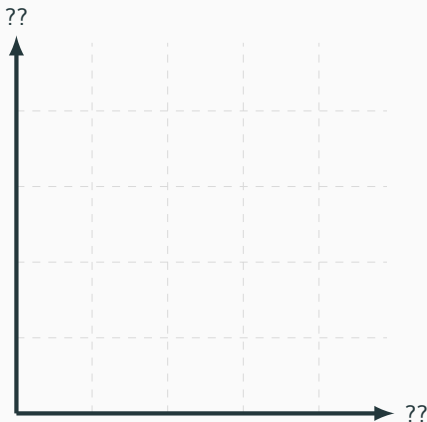
- Sketch a graph of driving speed against time on the highway.
- Sketch a graph of distance driven against time on the highway.
- Sketch a graph of the distance from my exit against time on the highway.
- Sketch a graph of the distance between cars against the distance driven on the highway.



One morning, first graph. . .

"As I drove down the highway this morning, at first traffic was fast and uncongested, then it crept nearly bumper-to-bumper until we passed an accident, after which traffic flow went back to normal until I exited."

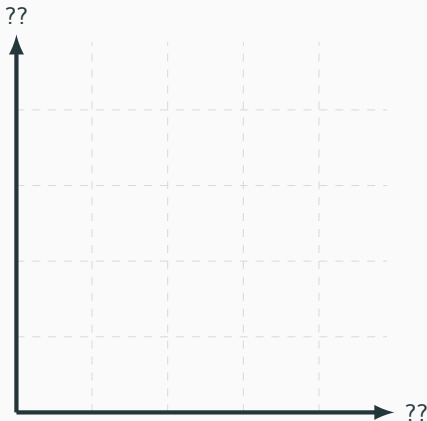
Sketch a graph of driving speed against time on the highway.



One morning, second graph. . .

"As I drove down the highway this morning, at first traffic was fast and uncongested, then it crept nearly bumper-to-bumper until we passed an accident, after which traffic flow went back to normal until I exited."

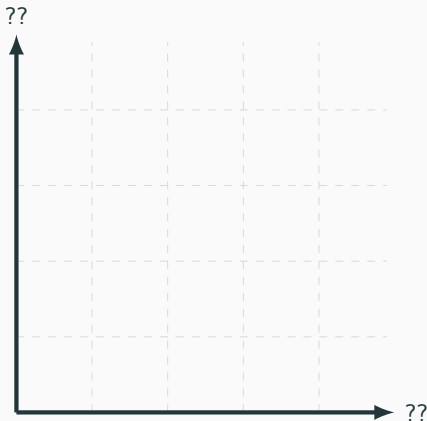
Sketch a graph of distance driven against time on the highway.



One morning, third graph. . .

"As I drove down the highway this morning, at first traffic was fast and uncongested, then it crept nearly bumper-to-bumper until we passed an accident, after which traffic flow went back to normal until I exited."

Sketch a graph of the distance from my exit against time on the highway.



One morning, fourth graph...

"As I drove down the highway this morning, at first traffic was fast and uncongested, then it crept nearly bumper-to-bumper until we passed an accident, after which traffic flow went back to normal until I exited."

Sketch a graph of the distance between cars against the distance driven on the highway.

